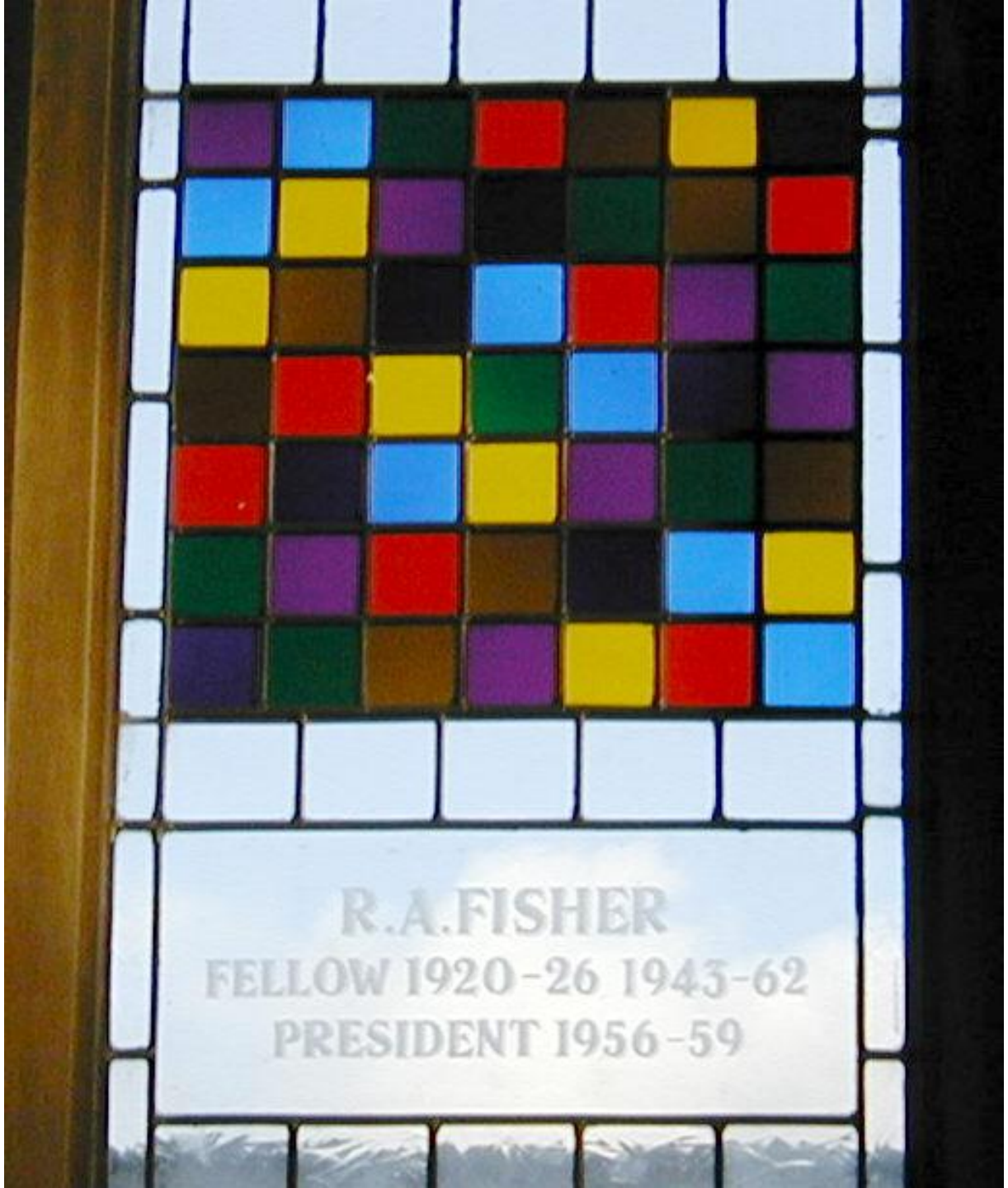
Comparing more than two things: ANOVA

Rolando Coto-Solano. Dartmouth College

LING10, January 2026


R.A.FISHER
FELLOW 1920-26 1943-62
PRESIDENT 1956-59

Types of ANOVAS

Independent one-way	One categorical variable 2 or more categories
Independent two-way (“factorial”)	2 or more categorical variables 2 or more categories
Repeated-Measures	Observations are paired

Two types of variation

What is the source of variation in a dataset?

x = typeOfWord
Categorical

y = Reaction Time
Numerical

nouns	within	$= \bar{x}_{nouns}$
verbs	within	$= \bar{x}_{verbs}$
adjectives	within	$= \bar{x}_{adjectives}$
pronouns	within	$= \bar{x}_{pronouns}$

between

Two types of variation

Within-group variation is the variation between the different people/items in each condition.

x = typeOfWord
Categorical

y = Reaction Time
Numerical

nouns	within	$= \bar{x}_{nouns}$
verbs	within	$= \bar{x}_{verbs}$
adjectives	within	$= \bar{x}_{adjectives}$
pronouns	within	$= \bar{x}_{pronouns}$

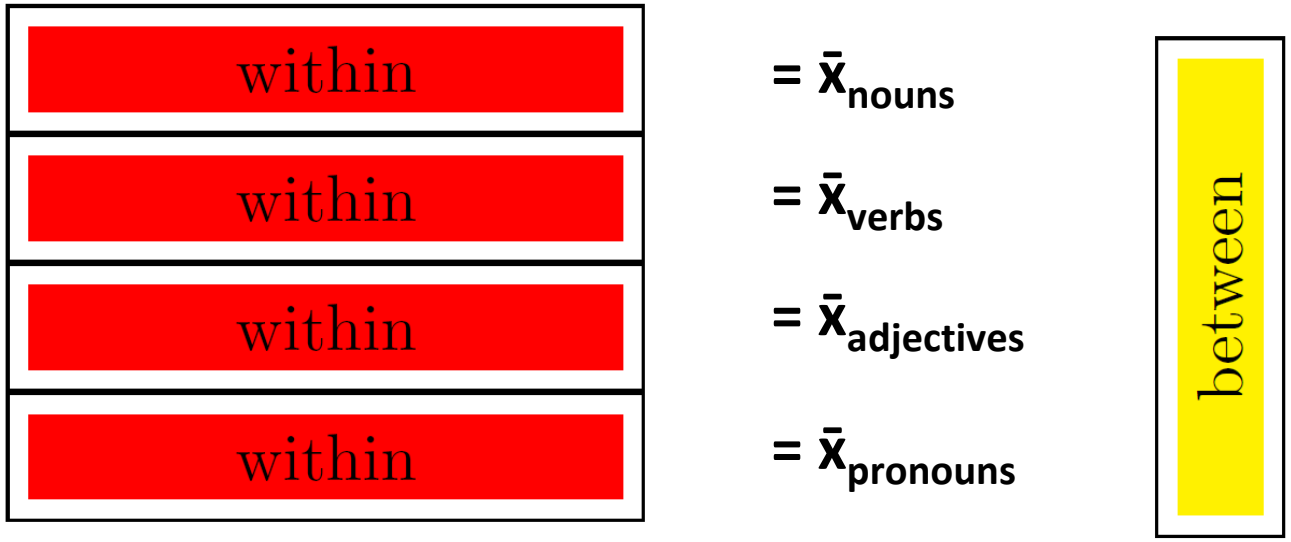
between

Two types of variation

Between-group variation is the variation in the sample means of the different categories in our x-variable.

x = typeOfWord
Categorical

y = Reaction Time
Numerical



Two types of variation

If the variation between categories is the **same** as the variation within, then we can't really say that there's a significant difference between the categories.

$$F = \frac{Var_{between}}{Var_{within}} \approx 1$$

x = typeOfWord
Categorical

y = Reaction Time
Numerical

nouns	within	$= \bar{x}_{nouns}$
verbs	within	$= \bar{x}_{verbs}$
adjectives	within	$= \bar{x}_{adjectives}$
pronouns	within	$= \bar{x}_{pronouns}$

between

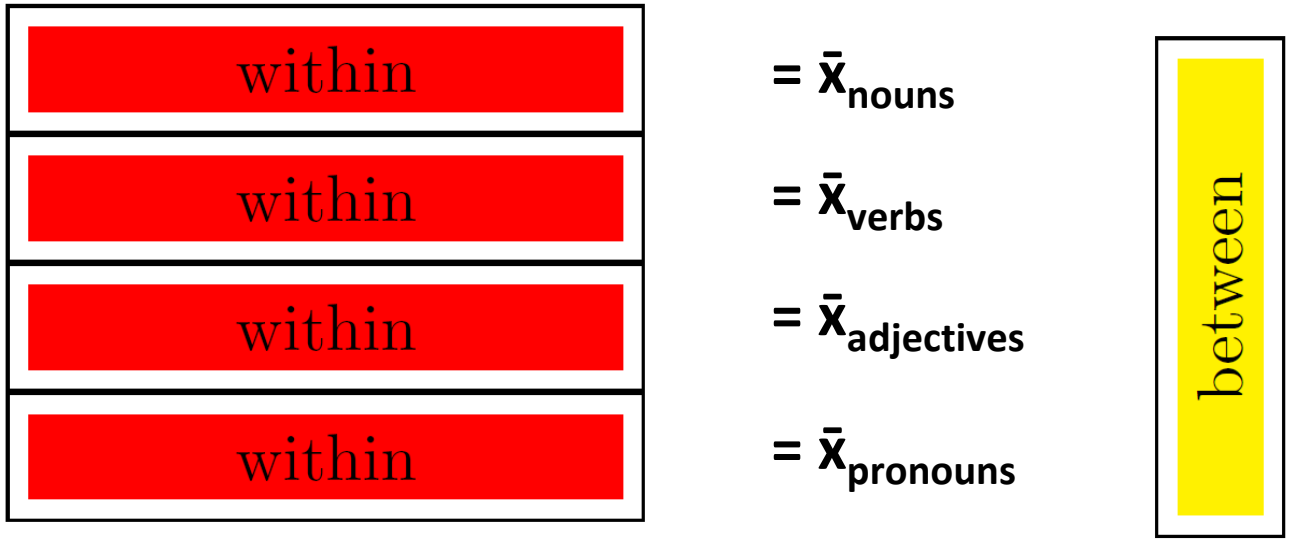
Two types of variation

If the variation between categories is **larger** than the variation within, then the sample means of the categories are distant from one another, and there might be a significant difference.

$$F = \frac{Var_{between}}{Var_{within}} > 1$$

x = typeOfWord
Categorical

y = Reaction Time
Numerical



Nicaraguan Sign Language

https://www.youtube.com/watch?v=U1I7IHY6xc4

Nicaraguan Sign Language case example

Dependent (y) = MannerPath (%, Numerical)

% MannerPath expressions
Percentage of expressions where children use separate words for manner of motion and path of motion (roll [MANNER] down [PATH])

Independent (x) = Generational group in the community (Categorical)

Early (Late 1970s, early 1980s)
Middle (Mid-late 1980s)
Late (1990s)

Test:

Whether $\bar{x}_{early}$, $\bar{x}_{middle}$, $\bar{x}_{late}$ are different
One-way ANOVA (because it's just one independent variable)

Nicaraguan Sign Language case example

Why not 3 separate t-tests?

($\bar{x}_{early}$ versus $\bar{x}_{middle}$)

($\bar{x}_{middle}$ versus $\bar{x}_{late}$)

($\bar{x}_{early}$ versus $\bar{x}_{late}$)

Nicaraguan Sign Language case example

Why not 3 separate t-tests?

($\bar{x}_{early}$ versus $\bar{x}_{middle}$)

($\bar{x}_{middle}$ versus $\bar{x}_{late}$)

($\bar{x}_{early}$ versus $\bar{x}_{late}$)

Because we'd be rolling the dice three times!

0.95 * 0.95 * 0.95 = 0.857

Probability of Type I error

(getting a significant result when the significance wasn't there)

Nicaraguan Sign Language case example

One-way ANOVA assumptions:

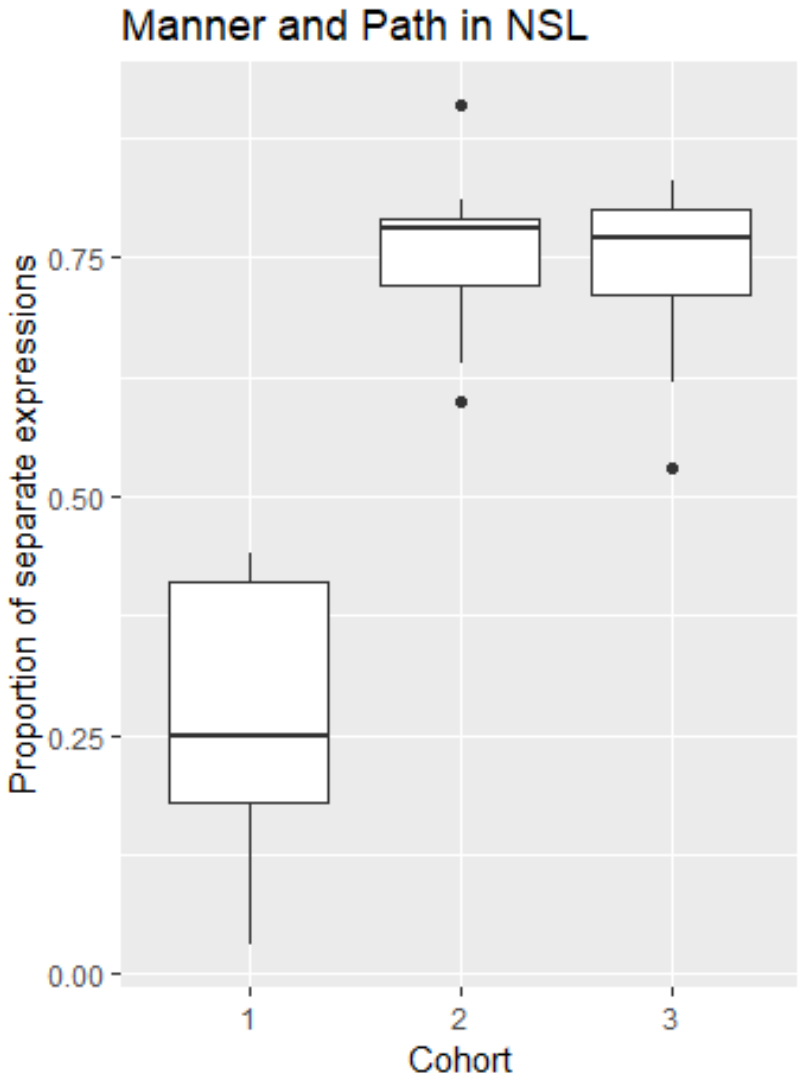
- The observations are independent (No people belong to two categories)

- The dependent/y/response variable is at least interval-scaled

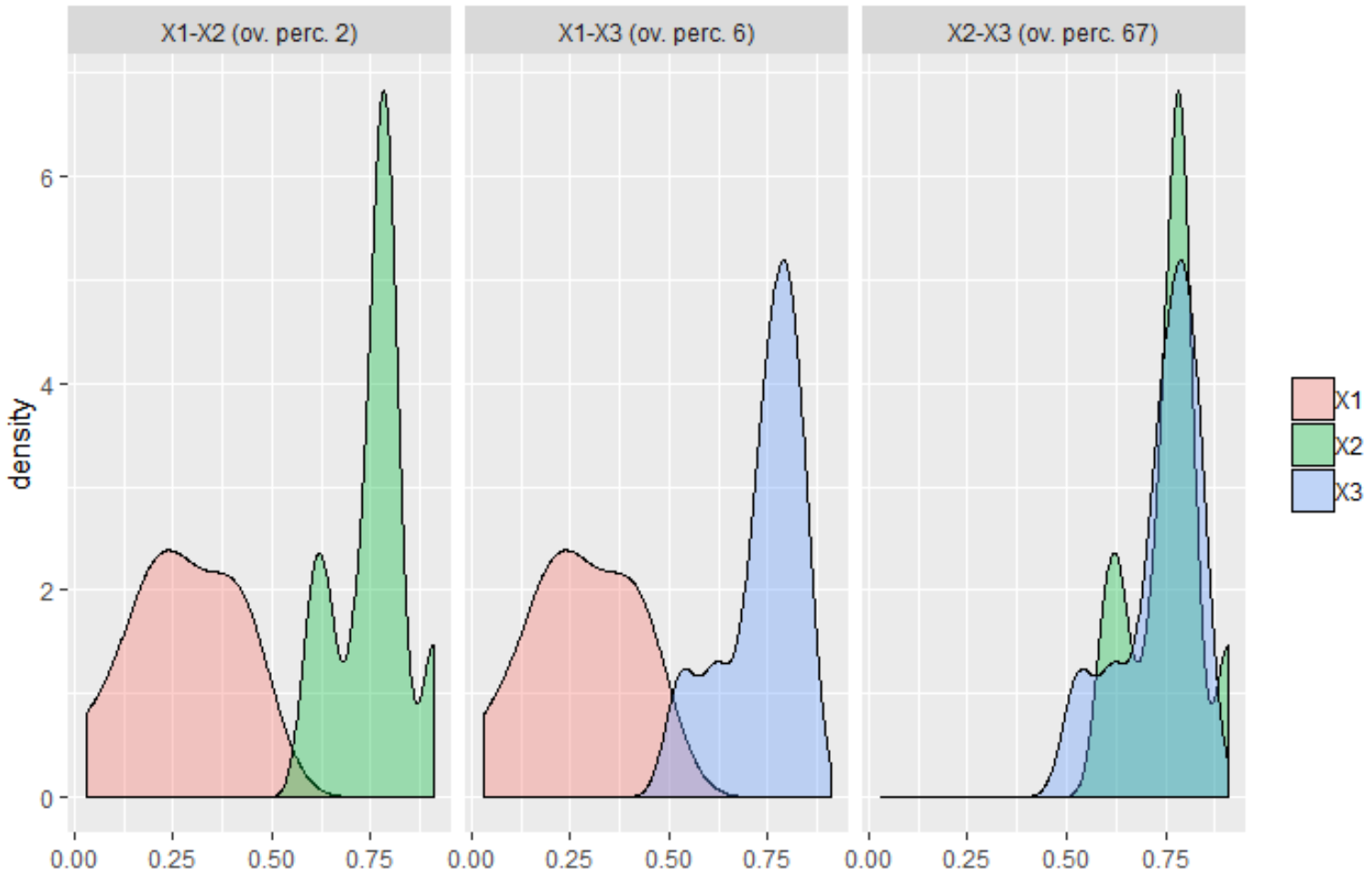
- The sample (for each category) is normally distributed, AND/OR the sample sizes are equal.

- The variance is roughly the same for all categories (is **homoscedastic**).

Nicaraguan sign case example



Early n=9 | Middle n=9 | Late n=9



Early - Middle | Early - Late | Middle - Late

Nicaraguan sign case example

A **balanced design** has an equal number of samples in each condition.

Because we have 9 children in each condition, this is balanced.

If we have an unbalanced design, ANOVA can still calculate it, but the worse the imbalance, the worse the results.

Nicaraguan sign case example

Normality assumption:

```
nsl1 = subset(nsl, Cohort==1)
nsl2 = subset(nsl, Cohort==2)
nsl3 = subset(nsl, Cohort==3)
shapiro.test(nsl1$MannerPath)
shapiro.test(nsl2$MannerPath)
shapiro.test(nsl3$MannerPath)
```

```
> shapiro.test(nsl1$MannerPath)
        Shapiro-Wilk normality test
data:  nsl1$MannerPath
W = 0.93805, p-value = 0.5615

> shapiro.test(nsl2$MannerPath)
        Shapiro-Wilk normality test
data:  nsl2$MannerPath
W = 0.93407, p-value = 0.521

> shapiro.test(nsl3$MannerPath)
        Shapiro-Wilk normality test
data:  nsl3$MannerPath
W = 0.87121, p-value = 0.1267
```

Nicaraguan sign case example

Normality assumption:

```
> aggregate(MannerPath ~ Cohort, data=NSL, function(x)
shapiro.test(x)$p.value)
  Cohort MannerPath
1      1  0.5614787
2      2  0.5210358
3      3  0.1266828
```

Nicaraguan sign case example

Same variances (homoscedasticity):

```
> leveneTest(MannerPath ~ Cohort, data=nsl)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  0.9207 0.4118
      24

> fligner.test(MannerPath ~ Cohort, data=nsl)

     Fligner-Killeen test of homogeneity of variances

data:  MannerPath by Cohort
Fligner-Killeen:med chi-squared = 2.3505, df = 2,
p-value = 0.3087
```

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

Nicaraguan sign case example

Two commands

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)
```

Independent variable

```
            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

We are calculating the mean of 3 categories.
Therefore, df=2

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

The sum (of the square) of the variation between categories.
(The sum of the variance)

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

The average variance between categories.

1.337 / 2 = 0.6684

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

The sum (of the square of) the variation within categories.
(the error, or "residual" variability).

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

The mean variance for each datapoint in the sample.

27 data points – 3 means = 24 df

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684  53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

$Var_{between} / Var_{within} = F$

$0.6684 / 0.0125 = 53.47$

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

Because there's a lot more variation between the categories than within them, our means must be very far apart.

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

Historically, asterisks have also been used to indicate significance.

Nicaraguan sign case example: Post-Hoc Comparisons

```
> TukeyHSD(anovaNSL)

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = MannerPath ~ Cohort, data = nsl)

$`Cohort`
           diff        lwr       upr   p adj
2-1  0.48222222  0.3505939 0.6138506 0.00000
3-1  0.46111111  0.3294828 0.5927394 0.00000
3-2 -0.02111111 -0.1527394 0.1105172 0.91568
```

Nicaraguan sign case example: Post-Hoc Comparisons

The input is the model we saved before.

```
> TukeyHSD(anovaNSL)

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = MannerPath ~ Cohort, data = nsl)

$`Cohort`
           diff        lwr       upr   p adj
2-1  0.48222222  0.3505939 0.6138506 0.00000
3-1  0.46111111  0.3294828 0.5927394 0.00000
3-2 -0.02111111 -0.1527394 0.1105172 0.91568
```

Nicaraguan sign case example: Post-Hoc Comparisons

```
> TukeyHSD(anovaNSL)

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = MannerPath ~ Cohort, data = nsl)

$`Cohort`
           diff        lwr       upr   p adj
2-1  0.48222222  0.3505939 0.6138506 0.00000
3-1  0.46111111  0.3294828 0.5927394 0.00000
3-2 -0.02111111 -0.1527394 0.1105172 0.91568
```

Confidence intervals when all the variance is taken into account

Nicaraguan sign case example: Post-Hoc Comparisons

```
> TukeyHSD(anovaNSL)

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = MannerPath ~ Cohort, data = nsl)

$`Cohort`
           diff        lwr       upr   p adj
2-1  0.48222222  0.3505939 0.6138506 0.00000
3-1  0.46111111  0.3294828 0.5927394 0.00000
3-2 -0.02111111 -0.1527394 0.1105172 0.91568
```

Differences in the sample means

Nicaraguan sign case example: Post-Hoc Comparisons

```
> TukeyHSD(anovaNSL)

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = MannerPath ~ Cohort, data = nsl)

$`Cohort`
           diff        lwr       upr    p adj
2-1  0.48222222  0.3505939 0.6138506 0.00000
3-1  0.46111111  0.3294828 0.5927394 0.00000
3-2 -0.02111111 -0.1527394 0.1105172 0.91568
```

95% confidence interval for the difference between groups.

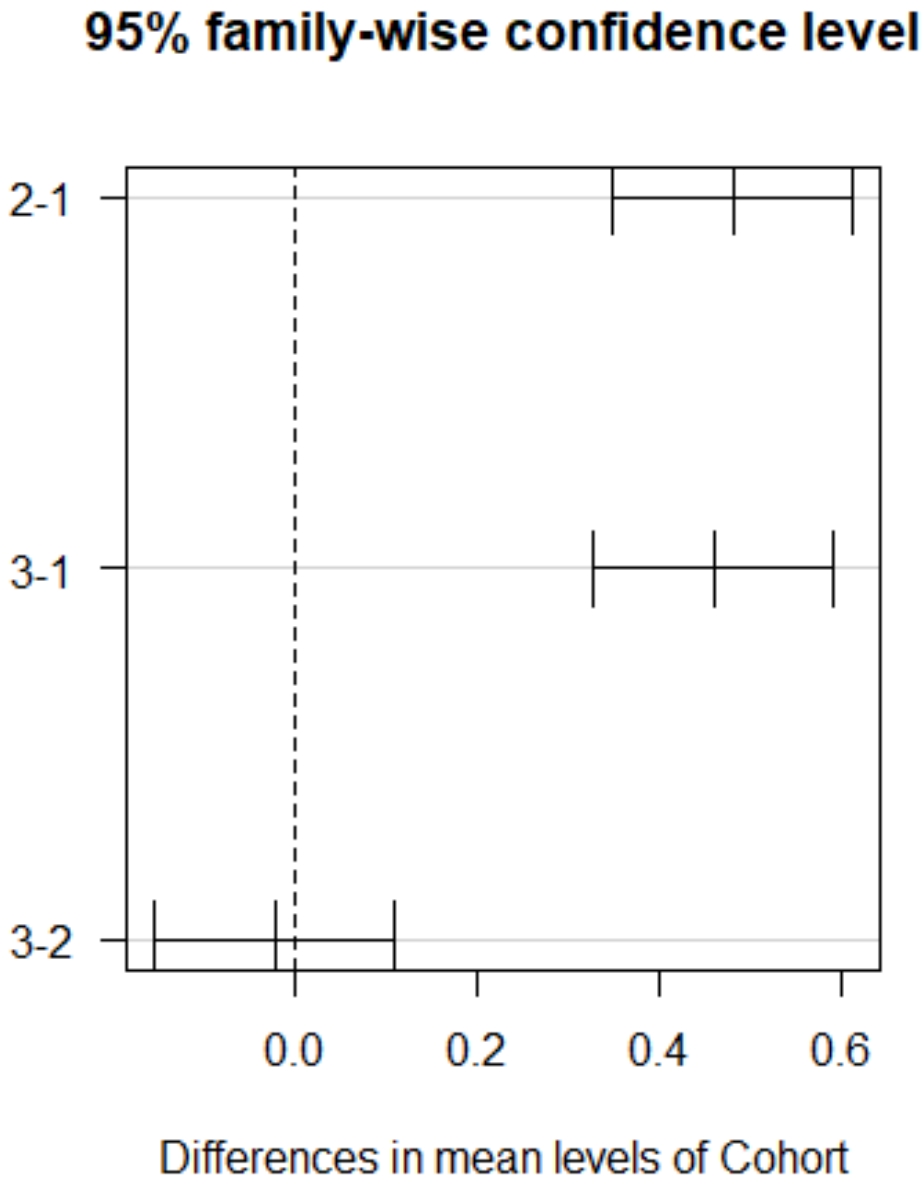
Nicaraguan sign case example: Post-Hoc Comparisons

```
> TukeyHSD(anovaNSL)

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = MannerPath ~ Cohort, data = nsl)

$`Cohort`
           diff        lwr       upr   p adj
2-1  0.48222222  0.3505939 0.6138506 0.00000
3-1  0.46111111  0.3294828 0.5927394 0.00000
3-2 -0.02111111 -0.1527394 0.1105172 0.91568
```

P-value adjusted for having throwing a dice so many times

Nicaraguan sign case example: Post-Hoc Comparisons

```
plot(TukeyHSD(anovaNSL),
las=1)
```



Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

The total (squared) variation in the dataset is 1.337+0.300 = 1.637

The `cohort` variable explains 1.337/1.637 = 0.817 = 81.7% of all of the variation in the data.

Nicaraguan sign case example

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

The total (squared) variation in the dataset is 1.337+0.300 = 1.637

The `cohort` variable explains 1.337/1.637 = 0.817 = 81.7% of all of the variation in the data.

There is still 0.300/1.637 = 18.3% of variation that remains unexplained.

Nicaraguan sign case example

```
> library('effectsize')
> eta_squared(anovaNSL)
# Effect Size for ANOVA

Parameter | Eta2 |        90% CI
-----------------------------------
Cohort    | 0.82 | [0.69, 0.88]
```

We call this measure eta-squared (η^2).
It's a measure of the **effect** of a variable.

$$\eta^2_{Cohort} = 0.82$$

Nicaraguan sign case example

```
> library('effectsize')
> eta_squared(anovaNSL)
# Effect Size for ANOVA

Parameter | Eta2 |        90% CI
-----------------------------------
Cohort    | 0.82 | [0.69, 0.88]
```

Is your effect "important"?

$0.01 < \eta^2 < 0.06$	Small
$0.06 < \eta^2 < 0.14$	Mid-size
$\eta^2 > 0.14$	Large

Guidelines for Effect Sizes: http://imaging.mrc-cbu.cam.ac.uk/statswiki/FAQ/effectSize

Summary so far

We have variation **between** groups and variation **within** groups.

If the variation between the groups is **larger** than the variation within, then **at least one mean must be different**.

If this happens, the ANOVA will give us a significant result.

We need to perform a post-hoc test to figure out which categories are actually significantly different from one other.

How to report an ANOVA

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

"A one-way between subjects ANOVA was conducted to compare the effect of Cohort on the percentage of expressions where manner and path are expressed through different words in Nicaraguan Sign Language. The cohort was divided in three generations: early (late 1970s-early 1980s), middle (mid-late 1980s) and late (1990s). The sample had a total of 27 children, 9 in each cohort."

How to report an ANOVA

```
> anovaNSL = aov(MannerPath ~ Cohort, data=nsl)
> summary(anovaNSL)

            Df Sum Sq Mean Sq F value   Pr(>F)
Cohort       2  1.337  0.6684   53.47 1.44e-09 ***
Residuals   24  0.300  0.0125
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

*"There was a significant effect of Cohort on the percentage of manner-path expressions ($F(2,24) = 53.5$, $p<0.001$, η^2=**?**)."*

How to report an ANOVA

```
> library('effectsize')
> eta_squared(anovaNSL)
# Effect Size for ANOVA

Parameter | Eta2 |       90% CI
-----------------------------------
Cohort    | 0.82 | [0.69, 0.88]
```

"There was a significant effect of Cohort on the percentage of manner-path expressions ($F(2,24) = 53.5$, $p<0.001$, $\eta^2=0.82$)."

How to report an ANOVA

```
> TukeyHSD(anovaNSL)

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = MannerPath ~ Cohort, data = nsl)

$`Cohort`
           diff        lwr       upr   p adj
2-1  0.48222222  0.3505939 0.6138506 0.00000
3-1  0.46111111  0.3294828 0.5927394 0.00000
3-2 -0.02111111 -0.1527394 0.1105172 0.91568
```

Post-hoc comparisons using the Tukey HSD test indicate that the mean for the early cohort (?, SD=?) is significantly different from the mean for the middle cohort (?, SD=?, p<0.0001) and the late cohort (?, SD=?, p<0.0001). There is no significant difference between the middle and the late cohort (p=0.92).

How to report an ANOVA

```
> aggregate(nsl$MannerPath, list(nsl$Cohort), mean)
  Group.1         x
1       1 0.2733333
2       2 0.7555556
3       3 0.7344444

> aggregate(nsl$MannerPath, list(nsl$Cohort), sd)
  Group.1          x
1       1 0.13711309
2       2 0.09234597
3       3 0.10088497
```

Post-hoc comparisons using the Tukey HSD test indicate that the mean for the early cohort (27%, SD=14%) is significantly different from the mean for the middle cohort (76%, SD=9%, p<0.0001) and the late cohort (73%, SD=10%, p<0.0001). There is no significant difference between the middle and the late cohort (p=0.92).

How to report an ANOVA

A one-way between subjects ANOVA was conducted to compare the effect of Cohort on the percentage of expressions where manner and path are expressed through different words in Nicaraguan Sign Language. The cohort was divided in three generations: early (late 1970s-early 1980s), middle (mid-late 1980s) and late (1990s). The cohort had a total of 27 children, 9 in each cohort.

"There was a significant effect of Cohort on the percentage of manner-path expressions ($F(2,24) = 53.5$, $p<0.001$, $\eta^2=0.82$). Post-hoc comparisons using the Tukey HSD test indicate that the mean for the early cohort (27%, SD=14%) is significantly different from the mean for the middle cohort (76%, SD=9%, $p<0.0001$) and the late cohort (73%, SD=10%, $p<0.0001$). There is no significant difference between the middle and the late cohort ($p=0.92$).

How to report an ANOVA

Between-group variation

A one-way between subjects ANOVA was conducted to compare the effect of Cohort on the percentage of expressions where manner and path are expressed through different words in Nicaraguan Sign Language. The cohort was divided in three generations: early (late 1970s-early 1980s), middle (mid-late 1980s) and late (1990s). The cohort had a total of 27 children, 9 in each cohort.

"There was a significant effect of Cohort on the percentage of manner-path expressions ($F(2,24) = 53.5$, $p<0.001$, $\eta^2=0.82$). Post-hoc comparisons using the Tukey HSD test indicate that the mean for the early cohort (27%, SD=14%) is significantly different from the mean for the middle cohort (76%, SD=9%, $p<0.0001$) and the late cohort (73%, SD=10%, $p<0.0001$). There is no significant difference between the middle and the late cohort ($p=0.92$).

How to report an ANOVA

Within-group variation

A one-way between subjects ANOVA was conducted to compare the effect of Cohort on the percentage of expressions where manner and path are expressed through different words in Nicaraguan Sign Language. The cohort was divided in three generations: early (late 1970s-early 1980s), middle (mid-late 1980s) and late (1990s). The sample had a total of 27 children, 9 in each cohort.

"There was a significant effect of Cohort on the percentage of manner-path expressions ($F(2,24) = 53.5$, $p<0.001$, $\eta^2=0.82$). Post-hoc comparisons using the Tukey HSD test indicate that the mean for the early cohort (27%, SD=14%) is significantly different from the mean for the middle cohort (76%, SD=9%, $p<0.0001$) and the late cohort (73%, SD=10%, $p<0.0001$). There is no significant difference between the middle and the late cohort ($p=0.92$).

Summary so far

ANOVA lets you compare more than 2 categories.

If the sample is normal and homoscedastic for each category, ANOVA can tell you how much variance comes from each of the categories (**eta-squared** , η^2).

Post-hoc tests (such as Tukey) help you compensate for throwing the dice too many times.

Independent Two-way ANOVA

Exercise: POS and RT

Independent Two-way ANOVA

We can expand our ANOVAs, so we can have **two independent** variables.

They both need to be **categorical**.

Nicaraguan Sign Language case example

Dependent (y) = Average of spatial modulation per verb (%, Numerical)

% Spatial modulations per verb
Spatial modulations: Signs are produced in a non-central position.
This indicates deictic and locative information.

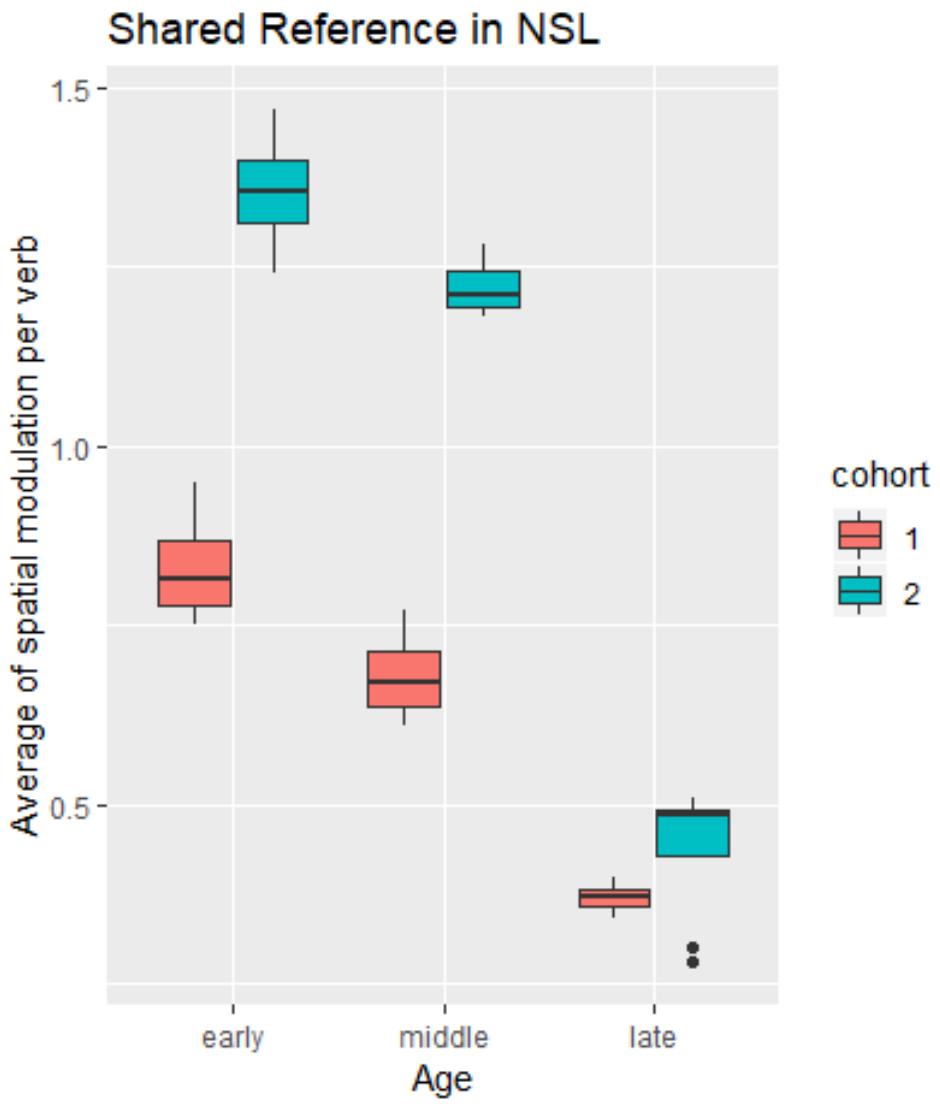
Independent 1 (x) = Age of children when they joined the school (Categorical)

Early (Before 6.5 years old)
Middle (Between 6.5 – 10 years old)
Late (After 10 years old)

Independent 2 (x) = Cohort: What decade did the person join:

1 and 2? (Early and middle?)

Independent Two-way ANOVA



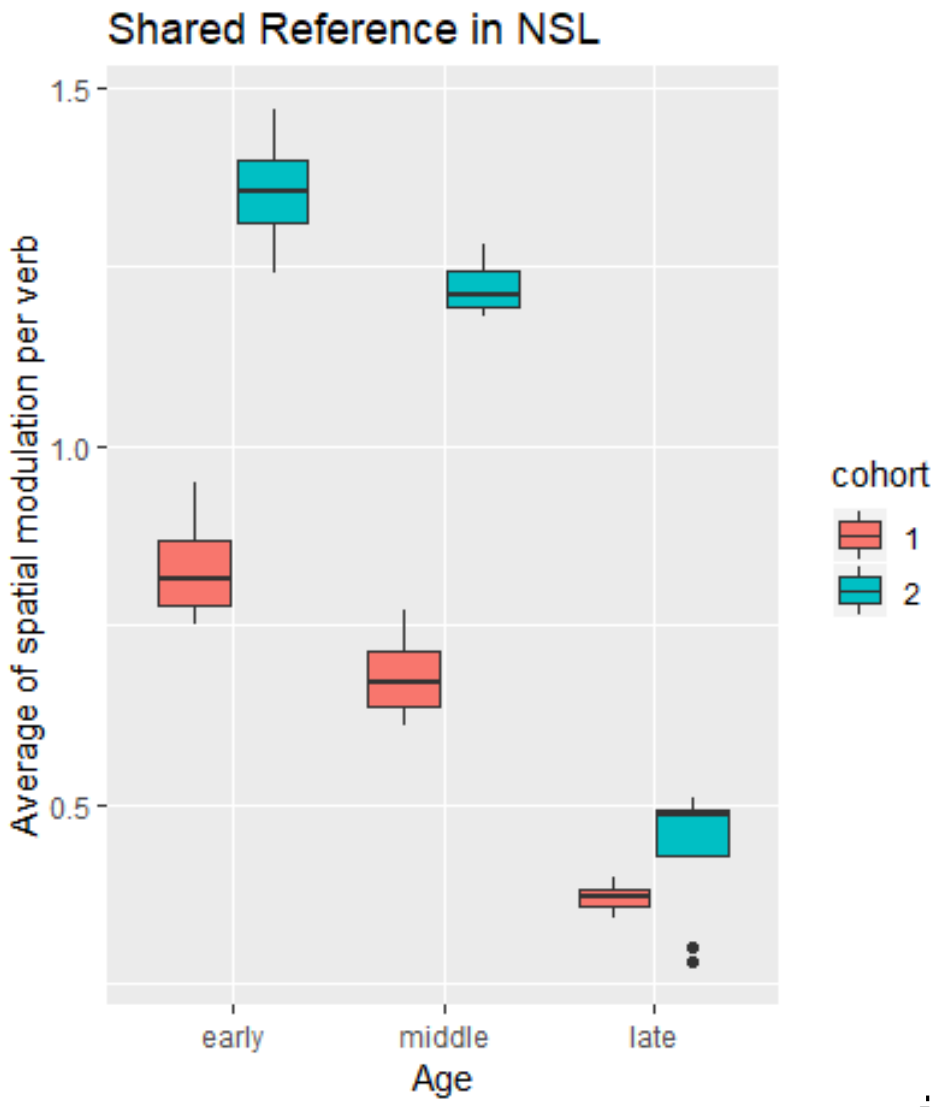
8 children per condition

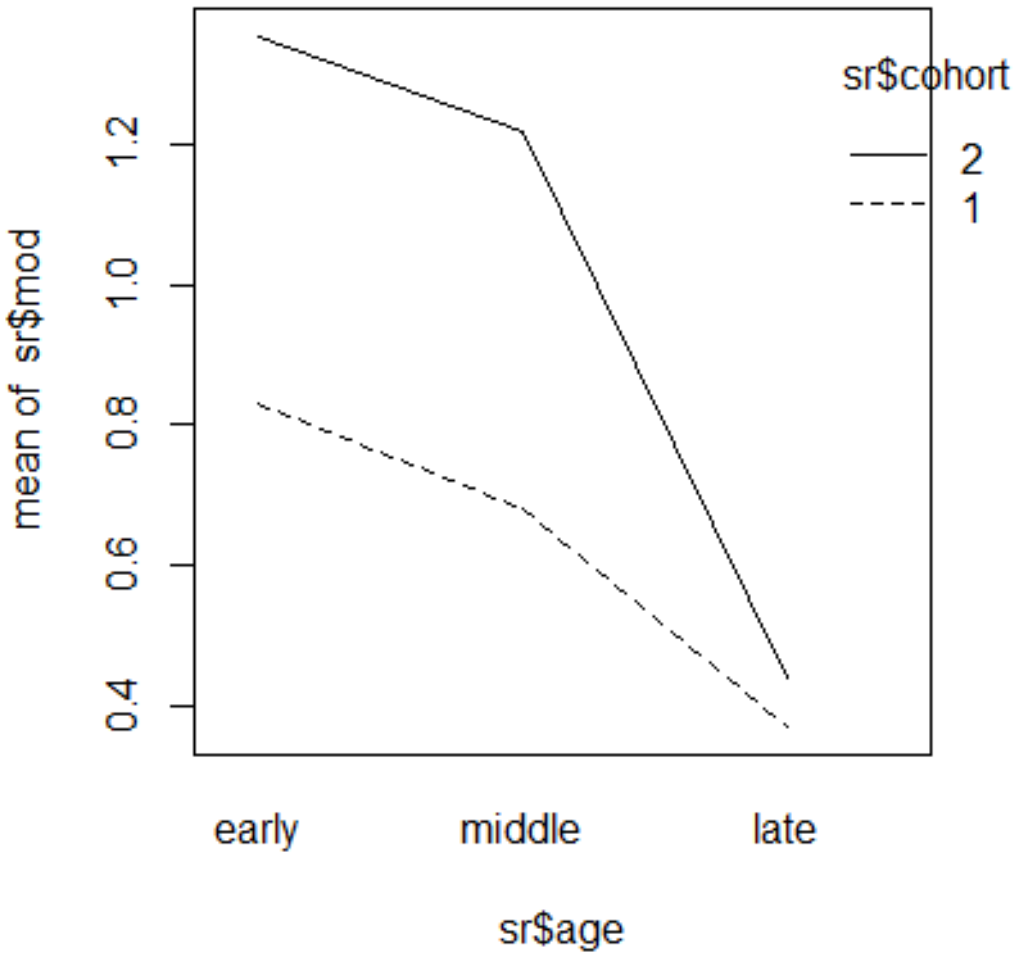
6 conditions (age,cohort):

Early-1
Middle-1
Late-1

Early-2
Middle-2
Late-2

Independent Two-way ANOVA



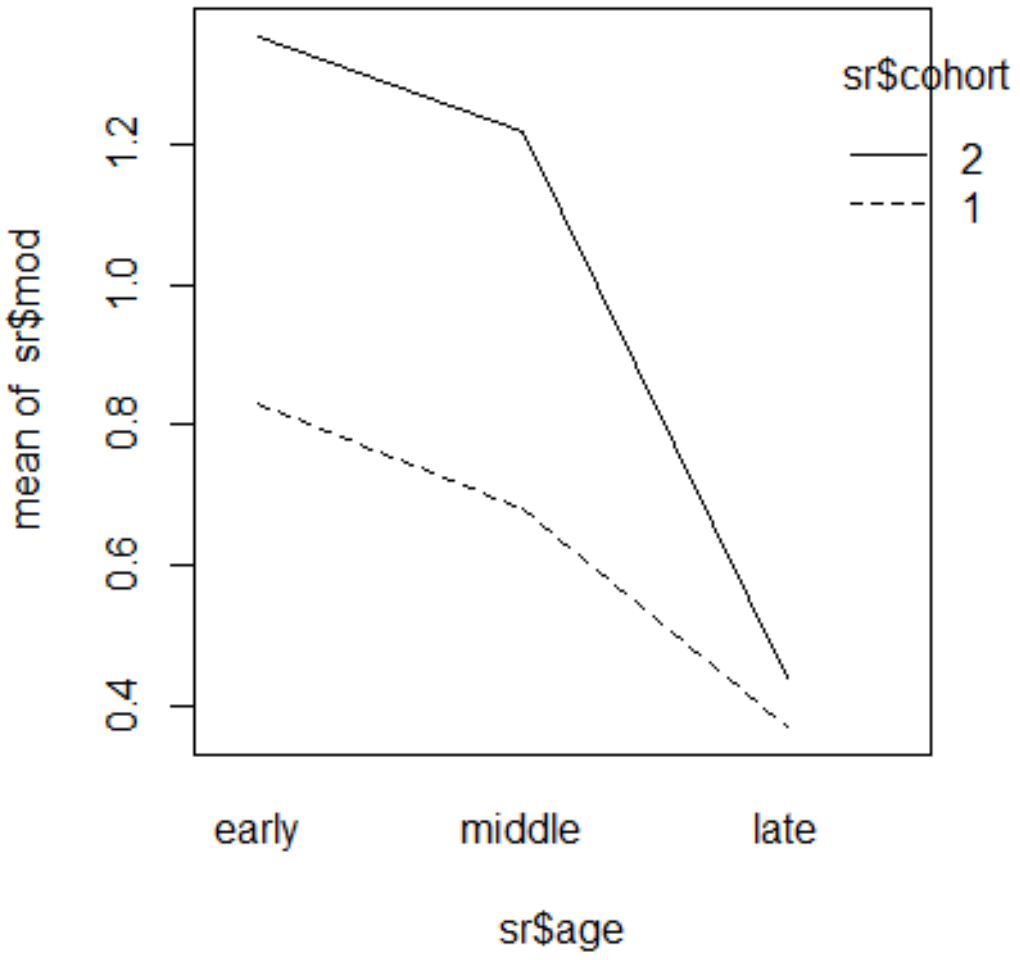



```
interaction.plot(sr$age, sr$cohort, sr$mod)
```

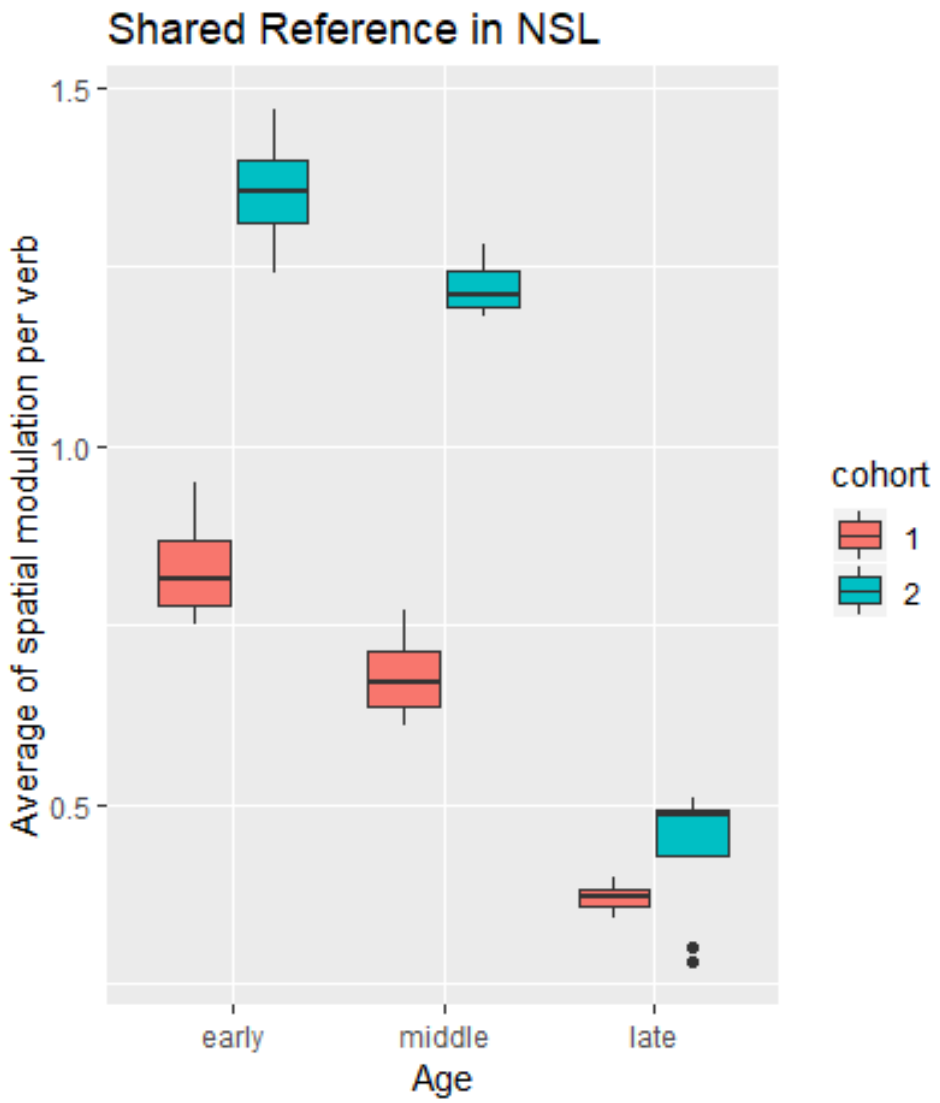
Independent Two-way ANOVA

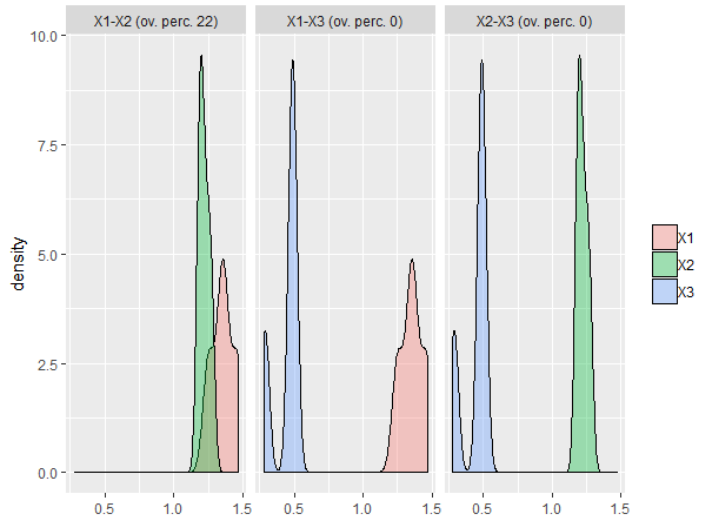
An **interaction** between variables happens when the combination of the two changes their behavior.

The effect of one variable depends on a second variable. (The effects aren't additive).

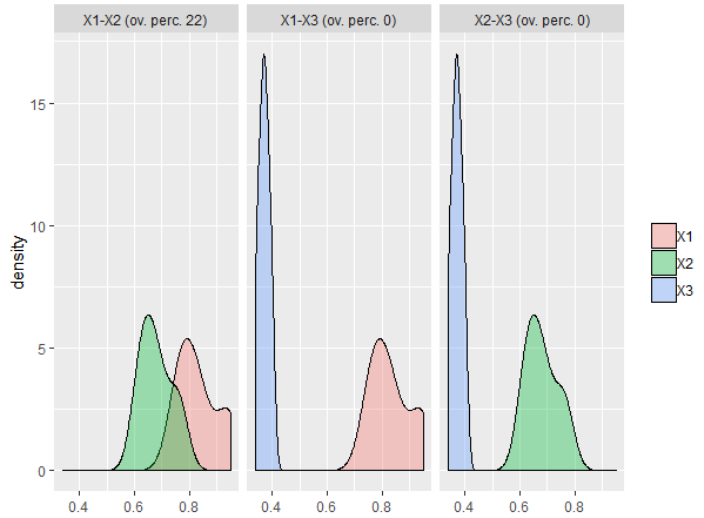


Independent Two-way ANOVA





C2



C1

Independent Two-way ANOVA

Normality:

```
> aggregate(mod ~ age + cohort, data = sr,
function(x) shapiro.test(x)$p.value)

     age cohort         mod
1  early      1 0.305382517
2 middle      1 0.611790881
3   late      1 0.975408765
4  early      2 0.710437680
5 middle      2 0.424433620
6   late      2 0.002067517
```

Independent Two-way ANOVA

Normality:

```
> aggregate(mod ~ age + cohort, data = sr,
function(x) shapiro.test(x)$p.value)
```

```
     age cohort         mod
1  early      1 0.305382517
2 middle      1 0.611790881
3   late      1 0.975408765
4  early      2 0.710437680
5 middle      2 0.424433620
6   late      2 0.002067517
```

We only have 8 children per condition, but most of the groups are normal. If we have a balanced design, ANOVA can deal with some deviations from normality. (ANOVA is **robust**).

Independent Two-way ANOVA

Same variance (homoscedasticity)

```
> leveneTest(mod ~ age * cohort, data=sr)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  5  1.1893 0.3308
      42
> fligner.test(mod ~ interaction(age,cohort), data=sr)

     Fligner-Killeen test of homogeneity of variances

data:  mod by interaction(age, cohort)
Fligner-Killeen:med chi-squared = 6.1292, df = 5,
p-value = 0.2939
```

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

             Df Sum Sq Mean Sq F value   Pr(>F)
age           2  4.224  2.1122  491.88  < 2e-16 ***
cohort        1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort    2  0.568  0.2840   66.13 1.05e-13 ***
Residuals    42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)
```

Main effects

```
              Df Sum Sq Mean Sq F value   Pr(>F)
age            2  4.224  2.1122  491.88  < 2e-16 ***
cohort         1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort     2  0.568  0.2840   66.13 1.05e-13 ***
Residuals     42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)
```

Interaction

```
              Df Sum Sq Mean Sq F value   Pr(>F)
age            2  4.224  2.1122  491.88  < 2e-16 ***
cohort         1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort     2  0.568  0.2840   66.13 1.05e-13 ***
Residuals     42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

If you have an interaction, this is your most important finding. Reporting only the main effects gives you a distorted view of the data.

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

             Df Sum Sq Mean Sq F value   Pr(>F)
age           2  4.224  2.1122  491.88  < 2e-16 ***
cohort        1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort    2  0.568  0.2840   66.13 1.05e-13 ***
Residuals    42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

We have three age groups, so df_{age} = 3-1 = 2

We have two cohorts, so df_{cohort} = 2-1 = 1

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

             Df Sum Sq Mean Sq F value   Pr(>F)
age           2  4.224  2.1122  491.88  < 2e-16 ***
cohort        1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort    2  0.568  0.2840   66.13 1.05e-13 ***
Residuals    42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

The degrees of freedom are $(df_{age} * df_{cohort}) = 2*1 = 2$

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

             Df Sum Sq Mean Sq F value   Pr(>F)
age           2  4.224  2.1122  491.88  < 2e-16 ***
cohort        1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort    2  0.568  0.2840   66.13 1.05e-13 ***
Residuals    42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

There are 48 samples.

We need to calculate 3*2 means. Therefore, we have $df_{residuals}$ = 48-6 = 42

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

             Df Sum Sq Mean Sq F value   Pr(>F)
age           2  4.224  2.1122  491.88  < 2e-16 ***
cohort        1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort    2  0.568  0.2840   66.13 1.05e-13 ***
Residuals    42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

$MeanSq_{age:cohort}$ = 0.568 / 2 = 0.284

$Var_{between(age:cohort)} / Var_{within}$ = $F_{age:cohort}$ = 0.284 / 0.0043 = 66.13

Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

             Df Sum Sq Mean Sq F value   Pr(>F)
age           2  4.224  2.1122  491.88  < 2e-16 ***
cohort        1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort    2  0.568  0.2840   66.13 1.05e-13 ***
Residuals    42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

All of the factors are significant, but because the interaction is significant, we know that the main effects don't tell the whole story.

Independent Two-way ANOVA

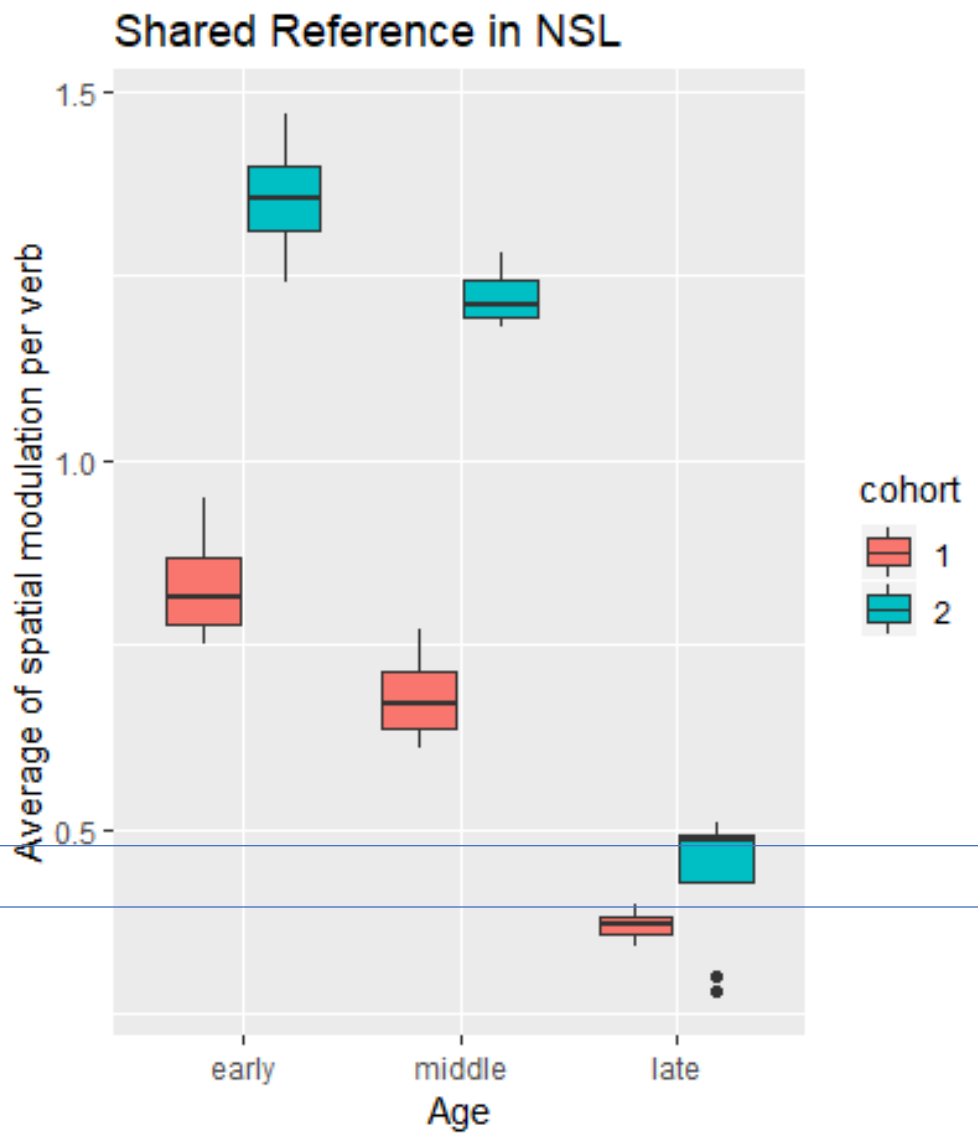
```
> TukeyHSD(sraov)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = mod ~ age * cohort, data = sr)

$`age`
                diff        lwr        upr p adj
middle-early -0.14375 -0.2000365 -0.0874635 6e-07
late-early   -0.68875 -0.7450365 -0.6324635 0e+00
late-middle  -0.54500 -0.6012865 -0.4887135 0e+00

$cohort
      diff       lwr       upr p adj
2-1 0.3775 0.3393248 0.4156752     0
```

Independent Two-way ANOVA

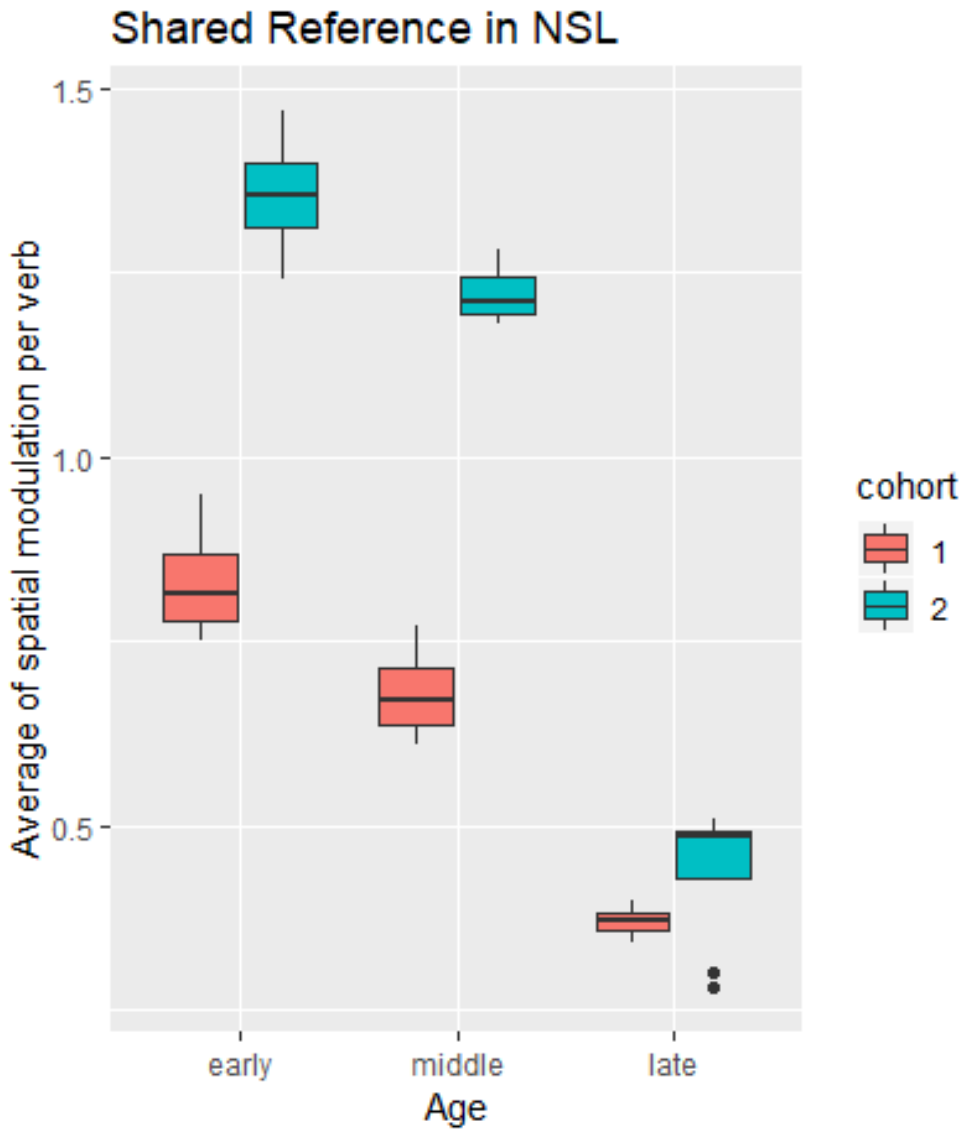
```
$`age:cohort`
                     diff         lwr         upr     p adj
middle:1-early:1  -0.1525 -0.25031016 -0.05468984 0.0004404
late:1-early:1    -0.4625 -0.56031016 -0.36468984 0.0000000
early:2-early:1    0.5225  0.42468984  0.62031016 0.0000000
middle:2-early:1   0.3875  0.28968984  0.48531016 0.0000000
late:2-early:1    -0.3925 -0.49031016 -0.29468984 0.0000000
late:1-middle:1   -0.3100 -0.40781016 -0.21218984 0.0000000
early:2-middle:1   0.6750  0.57718984  0.77281016 0.0000000
middle:2-middle:1  0.5400  0.44218984  0.63781016 0.0000000
late:2-middle:1   -0.2400 -0.33781016 -0.14218984 0.0000001
early:2-late:1     0.9850  0.88718984  1.08281016 0.0000000
middle:2-late:1    0.8500  0.75218984  0.94781016 0.0000000
late:2-late:1      0.0700 -0.02781016  0.16781016 0.2890363
middle:2-early:2  -0.1350 -0.23281016 -0.03718984 0.0022578
late:2-early:2    -0.9150 -1.01281016 -0.81718984 0.0000000
late:2-middle:2   -0.7800 -0.87781016 -0.68218984 0.0000000
```

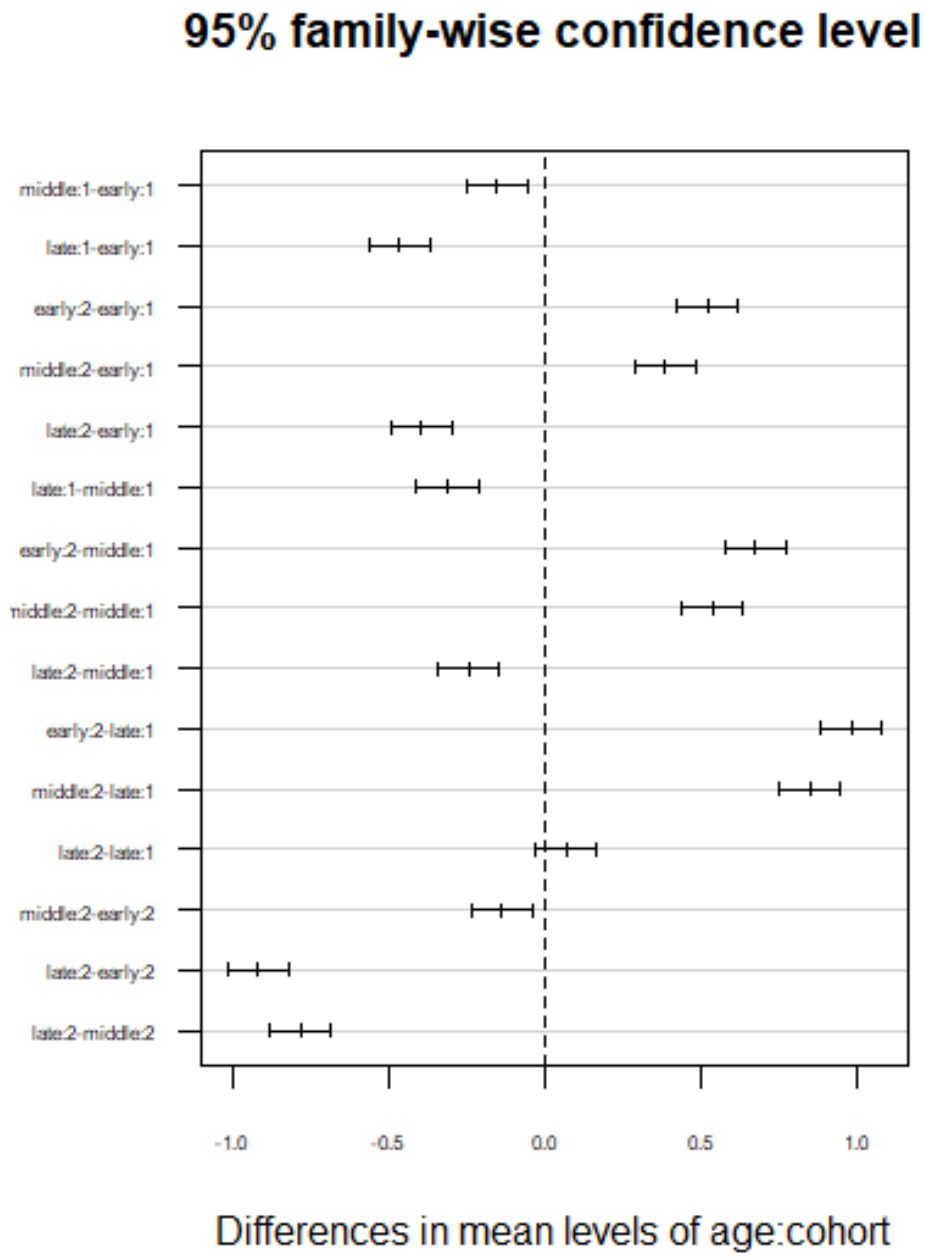
Independent Two-way ANOVA

```
$`age:cohort`

                          p adj
middle:1-early:1      0.0004404
late:1-early:1        0.0000000
early:2-early:1       0.0000000
middle:2-early:1      0.0000000
late:2-early:1        0.0000000
late:1-middle:1       0.0000000
early:2-middle:1      0.0000000
middle:2-middle:1     0.0000000
late:2-middle:1       0.0000001
early:2-late:1        0.0000000
middle:2-late:1       0.0000000
late:2-late:1         0.2890363
middle:2-early:2      0.0022578
late:2-early:2        0.0000000
late:2-middle:2       0.0000000
```



Independent Two-way ANOVA





Independent Two-way ANOVA

```
> eta_squared(sraov, partial="FALSE")
Parameter   | Eta2 |        90% CI
--------------------------------
age         | 0.63 | [0.47, 0.73]
cohort      | 0.26 | [0.09, 0.42]
age:cohort | 0.08 | [0.00, 0.22]
```

Age explains 63% of the variation.
Cohort expains 26% of the variation.
The interaction explains 8% of the variation.

There is 3% of variation that is still unexplained.

Independent Two-way ANOVA

```
> eta_squared(sraov, partial="FALSE")
Parameter   | Eta2 |        90% CI
--------------------------------
age         | 0.63 | [0.47, 0.73]
cohort      | 0.26 | [0.09, 0.42]
age:cohort  | 0.08 | [0.00, 0.22]
```

I am showing you η^2 because it's the most intuitive one. However, there are other measurements of effect size (partial-η^2, ω^2, partial-ω^2).

Age explains 63% of the variation.
Cohort expains 26% of the variation.
The interaction explains 8% of the variation.

There is 3% of variation that is still unexplained.

How to report an Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

            Df Sum Sq Mean Sq F value   Pr(>F)
age          2  4.224  2.1122  491.88  < 2e-16 ***
cohort       1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort   2  0.568  0.2840   66.13 1.05e-13 ***
Residuals   42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

A two-way, between-subjects ANOVA was conducted to compare the effect of Cohort and Age on the number of "shared reference" expressions per verb in NSL. The age has three levels: early (children that entered school before they were 6.5 years old), middle (children that entered school when they were between 6.5 and 10 years old) and late (children that entered school after they were 10 years old). Cohort has two levels: 1 and 2. There are a total of 48 children, 8 for each of the six combinations of age and cohort.

How to report an Independent Two-way ANOVA

```
> sraov = aov(mod ~ age*cohort, data=sr)
> summary(sraov)

            Df Sum Sq Mean Sq F value   Pr(>F)
age          2  4.224  2.1122  491.88  < 2e-16 ***
cohort       1  1.710  1.7101  398.24  < 2e-16 ***
age:cohort   2  0.568  0.2840   66.13 1.05e-13 ***
Residuals   42  0.180  0.0043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

```
> eta_squared(sraov, partial="FALSE")

Parameter   | Eta2 |       90% CI
--------------------------------
age         | 0.63 | [0.47, 0.73]
cohort      | 0.26 | [0.09, 0.42]
age:cohort  | 0.08 | [0.00, 0.22]
```

There is a significant effect in the interaction of age and cohort ($F(2,42)=66.1$, $p<0.001$, $\eta^2=0.08$). There are also significant main effects for age ($F(2,42)=491$, $p<0.0001$, $\eta^2=0.63$) and Cohort ($F(1,42)=398$, $p<0.0001$, $\eta^2=0.26$).

How to report an Independent Two-way ANOVA

```
> aggregate(mod ~ age + cohort,
data=sr, FUN=mean)
     age cohort    mod
1  early      1 0.8325
2 middle      1 0.6800
3   late      1 0.3700
4  early      2 1.3550
5 middle      2 1.2200
6   late      2 0.4400
```

```
> aggregate(mod ~ age + cohort,
data=sr, FUN=sd)
     age cohort        mod
1  early      1 0.07382412
2 middle      1 0.05707138
3   late      1 0.02000000
4  early      2 0.08088793
5 middle      2 0.03703280
6   late      2 0.09350325
```

A post-hoc comparison using the Tukey HSD test reveals further details about the interaction: ***In the first cohort, the children who entered school at an early age use more shared-reference expressions than children who entered at a middle age*** *(*$mean_{early:1}$ *= 0.83±0.07,* $mean_{middle:1}$*=0.68±0.06, p<0.0005)* ***or a late age*** *(*$mean_{late:1}$ *= 0.37±0.02, p<0.0001).* ***This is also true for the second cohort, who uses the shared reference even more*** *(*$mean_{early:2}$ *= 1.35±0.08,* $mean_{middle:2}$*=1.22±0.04,* $mean_{late:2}$ *= 0.44%±0.09, early-middle: p<0.005, early-late: p<0.0001).*

How to report an Independent Two-way ANOVA

```
> aggregate(mod ~ age + cohort,
data=sr, FUN=mean)
     age cohort    mod
1  early      1 0.8325
2 middle      1 0.6800
3   late      1 0.3700
4  early      2 1.3550
5 middle      2 1.2200
6   late      2 0.4400
```

```
> aggregate(mod ~ age + cohort,
data=sr, FUN=sd)
     age cohort        mod
1  early      1 0.07382412
2 middle      1 0.05707138
3   late      1 0.02000000
4  early      2 0.08088793
5 middle      2 0.03703280
6   late      2 0.09350325
```

A post-hoc comparison using the Tukey HSD test reveals further details about the interaction: In the first cohort, the children who entered school at an early age use more shared-reference expressions than children who entered at a middle age ($mean_{early:1}$ = 0.83±0.07, $mean_{middle:1}$=0.68±0.06, p<0.0005) or a late age ($mean_{late:1}$ = 0.37±0.02, p<0.0001). This is also true for the second cohort, who uses the shared reference even more ($mean_{early:2}$ = 1.35±0.08, $mean_{middle:2}$=1.22±0.04, $mean_{late:2}$ = 0.44%±0.09, early-middle: p<0.005, early-late: p<0.0001). However, there is an important interaction: ***Children who entered school in the late age, after they became ten years old, were much lower in their use of shared-reference*** *(Cohort1: 0.37±0.02 versus Cohort2: 0.44%±0.09). There is no significant difference between them (p=0.29).*

How to report an Independent Two-way ANOVA

```
> aggregate(mod ~ age + cohort,
data=sr, FUN=mean)
     age cohort    mod
1  early      1 0.8325
2 middle      1 0.6800
3   late      1 0.3700
4  early      2 1.3550
5 middle      2 1.2200
6   late      2 0.4400
```

```
> aggregate(mod ~ age + cohort,
data=sr, FUN=sd)
     age cohort        mod
1  early      1 0.07382412
2 middle      1 0.05707138
3   late      1 0.02000000
4  early      2 0.08088793
5 middle      2 0.03703280
6   late      2 0.09350325
```

A post-hoc comparison using the Tukey HSD test reveals further details about the interaction: In the first cohort, the children who entered school at an early age use more shared-reference expressions than children who entered at a middle age ($mean_{early:1}$ *= 0.83±0.07,* $mean_{middle:1}$*=0.68±0.06, p<0.0005) or a late age (*$mean_{late:1}$ *= 0.37±0.02, p<0.0001). This is also true for the second cohort, who uses the shared reference even more (*$mean_{early:2}$ *= 1.35±0.08,* $mean_{middle:2}$*=1.22±0.04,* $mean_{late:2}$ *= 0.44%±0.09, early-middle: p<0.005, early-late: p<0.0001). However, there is an important interaction: Children who entered school in the late age, after they became ten years old, were much lower in their use of shared-reference (Cohort1: 0.37±0.02 versus Cohort2: 0.44%±0.09). There is no significant difference between them (p=0.29).*

Footnote:The number after '±' is the SD.

How to report an Independent Two-way ANOVA

A two-way, between-subjects ANOVA was conducted to compare the effect of Cohort and Age on the number of "shared reference" expressions per verb in NSL. The age has three levels: early (children that entered school before they were 6.5 years old), middle (children that entered school when they were between 6.5 and 10 years old) and late (children that entered school after they were 10 years old). Cohort has two levels: 1 and 2. There are a total of 48 children, 8 for each of the six combinations of age and cohort.

There is a significant effect in the interaction of age and cohort ($F(2,42)=66.1$, $p<0.001$, $\eta^2=0.08$). There are also significant main effects for age ($F(2,42)=491$, $p<0.0001$, $\eta^2=0.63$) and Cohort ($F(1,42)=398$, $p<0.0001$, $\eta^2=0.26$).

A post-hoc comparison using the Tukey HSD test reveals further details about the interaction: In the first cohort, the children who entered school at an early age use more shared-reference expressions than children who entered at a middle age ($mean_{early:1} = 0.83\pm0.07$, $mean_{middle:1}=0.68\pm0.06$, $p<0.0005$) or a late age ($mean_{late:1} = 0.37\pm0.02$, $p<0.0001$). This is also true for the second cohort, who uses the shared reference even more ($mean_{early:2} = 1.35\pm0.08$, $mean_{middle:2}=1.22\pm0.04$, $mean_{late:2} = 0.44\%\pm0.09$, early-middle: $p<0.005$, early-late: $p<0.0001$). However, there is an important interaction: Children who entered school in the late age, after they became ten years old, were much lower in their use of shared-reference, regardless of cohort (Cohort1: 0.37±0.02 versus Cohort2: 0.44%±0.09). There is no significant difference between them ($p=0.29$).

Footnote:The number after '±' is the SD.

How to report an Independent Two-way ANOVA

A two-way, between-subjects ANOVA was conducted to compare the effect of Cohort and Age on the number of "shared reference" expressions per verb in NSL. The age has three levels: early (children that entered school before they were 6.5 years old), middle (children that entered school when they were between 6.5 and 10 years old) and late (children that entered school after they were 10 years old). Cohort has two levels: 1 and 2. There are a total of 48 children, 8 for each of the six combinations of age and cohort.

There is a significant effect in the interaction of age and cohort ($F(2,42)=66.1$, $p<0.001$, $\eta^2=0.08$). There are also significant main effects for age ($F(2,42)=491$, $p<0.0001$, $\eta^2=0.63$) and Cohort ($F(1,42)=398$, $p<0.0001$, $\eta^2=0.26$). A post-hoc comparison using the Tukey HSD test reveals further details about the interaction: In the first cohort, the children who entered school at an early age use more shared-reference expressions than children who entered at a middle age (0.83 versus 0.68, $p<0.0005$) or a late age (0.37, $p<0.0001$). This is also true for the second cohort, who uses the shared reference even more (early: 1.35, middle: 1.22, late: 0.44, early-middle: $p<0.005$, early-late: $p<0.0001$). However, there is an important interaction: Children who entered school in the late age, after they became ten years old, were much lower in their use of shared-reference, regardless of cohort (Cohort1: 0.37 versus Cohort2: 0.44%). There is no significant difference between them ($p=0.29$)

	Cohort1	Cohort2
Early	0.83 ± 0.07	1.35 ± 0.08
Middle	0.68 ± 0.06	1.22 ± 0.04
Late	0.37 ± 0.02	0.44 ± 0.09

'±' indicates standard deviation

Summary so far

We have two types of ANOVAs:

One-way between subjects	One independent variable, categorical (2 or more categories)
Two-way between subjects	Two or more independent variables, categorical (2 or more categories in each of them)

We can figure out how much **variance** is explained by each variable (**eta-squared**).

We can figure out whether there are **significant difference in the means** of categories (**p-value**)

We can figure out if there are **interactions** between the variables.

Independent Two-way ANOVA

Exercise: French Consonants

Between subjects and within subjects

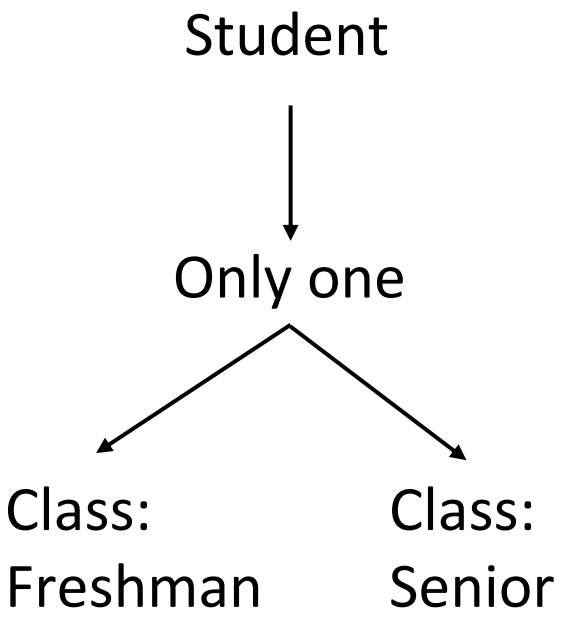
In some variables, each participant can only have one value.
These are **between-subjects** variables.

Student	Class	VOT
S01	Freshman	110
S02	Freshman	115
S03	Freshman	120
S04	Senior	118
S05	Senior	124
S06	Senior	116

Between subjects and within subjects

In some variables, each participant can only have one value.
These are **between-subjects** variables.

Student	Class	VOT
S01	Freshman	110
S02	Freshman	115
S03	Freshman	120
S04	Senior	118
S05	Senior	124
S06	Senior	116



Between subjects and within subjects

In some variables, the participant can take more than one value.
These are **within-subjects** variables.

Student	Class	Test	VOT
S01	Freshman	Before	110
S01	Freshman	After	116
S02	Freshman	Before	115
S02	Freshman	After	121
S03	Freshman	Before	120
S03	Freshman	After	119
S04	Senior	Before	118
S04	Senior	After	125

Between subjects and within subjects

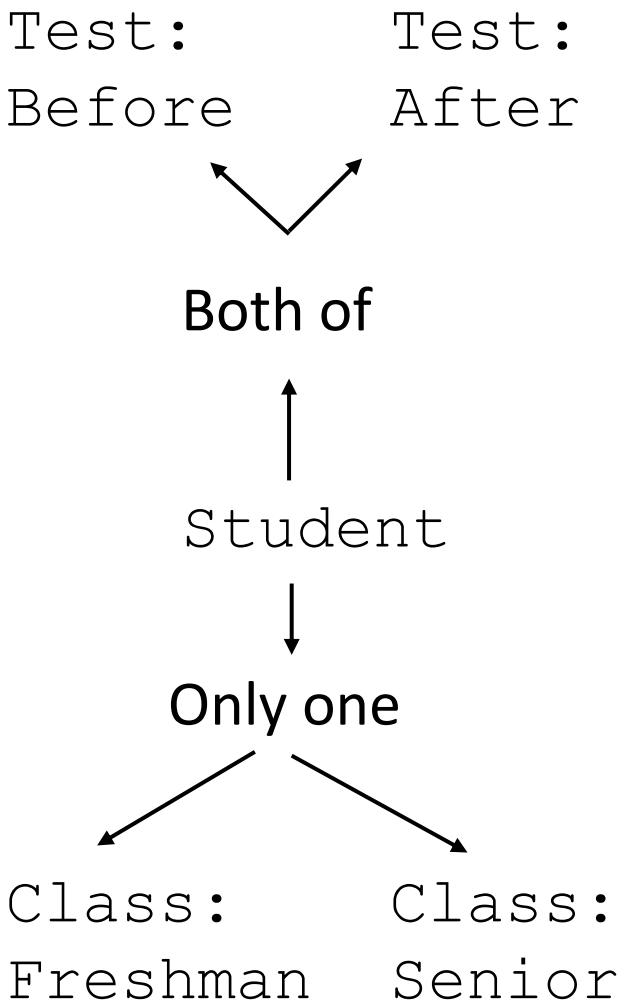
In some variables, the participant can take more than one value.
These are **within-subjects** variables.

Student	Class	Test	VOT
S01	Freshman	Before	110
S01	Freshman	After	116
S02	Freshman	Before	115
S02	Freshman	After	121
S03	Freshman	Before	120
S03	Freshman	After	119
S04	Senior	Before	118
S04	Senior	After	125

Between subjects and within subjects

In some variables, the participant can take more than one value.
These are **within-subjects** variables.

Student	Class	Test	VOT
S01	Freshman	Before	110
S01	Freshman	After	116
S02	Freshman	Before	115
S02	Freshman	After	121
S03	Freshman	Before	120
S03	Freshman	After	119
S04	Senior	Before	118
S04	Senior	After	125



Repeated Measures ANOVA

An ANOVA with a within-subjects variable is a **Repeated Measures ANOVA**.

Repeated Measures ANOVA

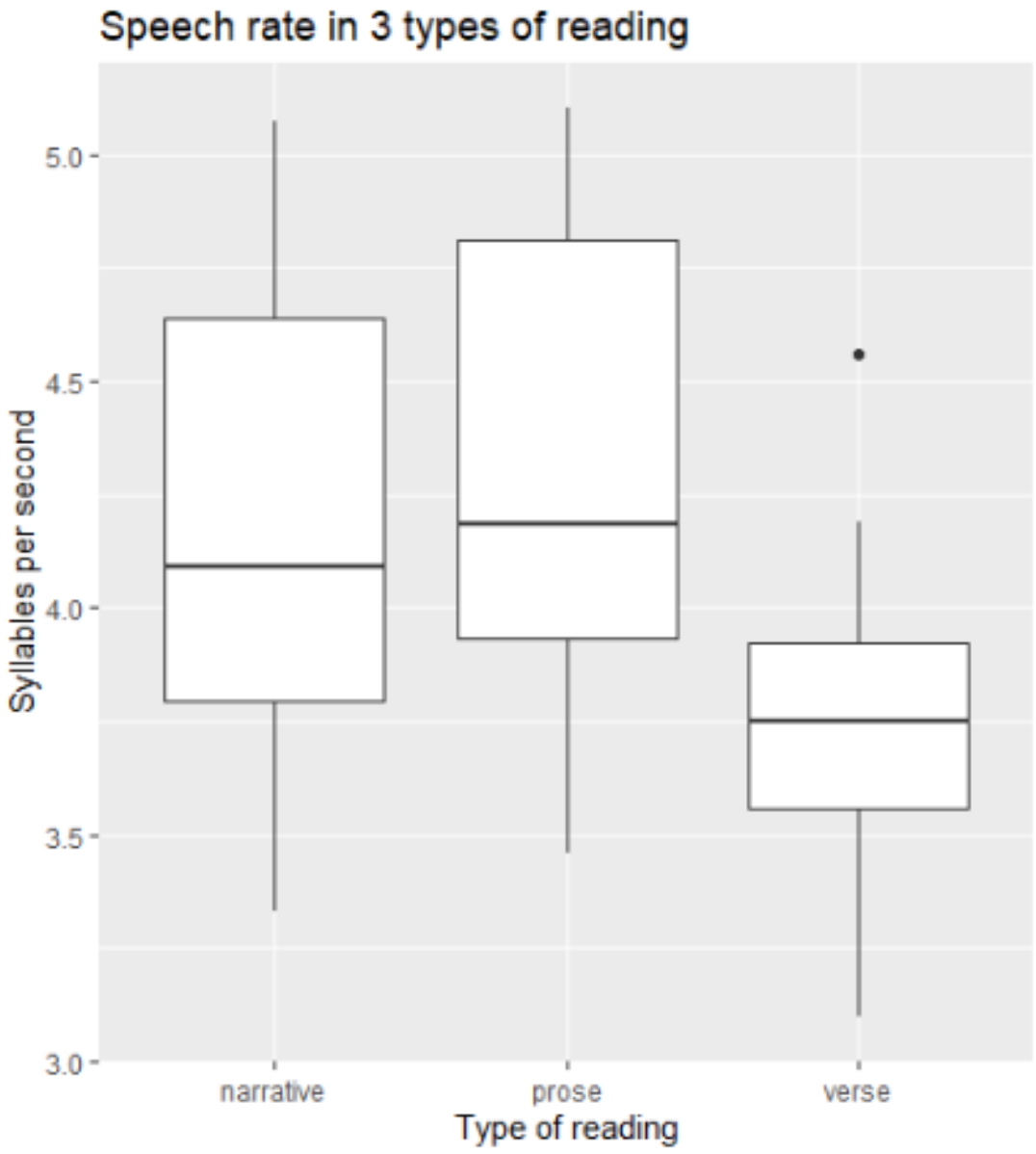
Students at a university in Moscow (High School of Economics) read several passages from different books. Each person reads from:

`type`: (1) General prose (2) Narrative prose (storytelling) (3) Verse (poetry)

We measure the `syllsPerSecond` (the speech rate) for 20 different people:

Student	type	syllsPerSecond
S01	Prose	3.5
S01	Narr	3.6
S01	Verse	3.4
S02	Prose	4.4
S02	Narr	3.8
S02	Verse	4.6
...		

Repeated Measures ANOVA



Repeated Measures ANOVA

```
> aggregate(syllsPerSec ~ type, data=readingRate, function(x)
shapiro.test(x)$p.value)
       type syllsPerSec
1 narrative   0.2894951
2     prose   0.2540872
3     verse   0.9468102
```

```
> leveneTest(syllsPerSec ~ type, data=readingRate)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  2.4223 0.0978 .
      57
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Repeated Measures ANOVA

```
> rr.aov = aov(syllsPerSec ~ type + Error(student/type),
data=readingRate)
> summary(rr.aov)
```

Student	type	syllsPerSecond
S01	Prose	3.5
S01	Narr	3.6
S01	Verse	3.4
S02	Prose	4.4
S02	Narr	3.8
S02	Verse	4.6
...		

Repeated Measures ANOVA

```
> rr.aov = aov(syllsPerSec ~ type + Error(student/type),
data=readingRate)
> summary(rr.aov)

Error: student
          Df Sum Sq Mean Sq F value Pr(>F)
Residuals 19  3.482  0.1833

Error: student:type
          Df Sum Sq Mean Sq F value  Pr(>F)
type       2  3.600  1.8002   6.975 0.00263 **
Residuals 38  9.807  0.2581
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Repeated Measures ANOVA

```
> rr.aov = aov(syllsPerSec ~ type + Error(student/type),
data=readingRate)
> summary(rr.aov)

Error: student
          Df Sum Sq Mean Sq F value Pr(>F)
Residuals 19  3.482  0.1833

Error: student:type
          Df Sum Sq Mean Sq F value  Pr(>F)
type       2  3.600  1.8002   6.975 0.00263 **
Residuals 38  9.807  0.2581
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Repeated Measures ANOVA

```
> eta_squared(rr.aov, partial=FALSE)
Group          | Parameter | Eta2 |        90% CI
-----------------------------------------------
student:type |      type | 0.21 | [0.03, 0.38]
```

Repeated Measures ANOVA

Short story:

There are no good Post-Hoc tests for repeated measures ANOVA! Within-subjects variables can be ordinal or categorical, which changes the analysis. So we'll stop here.

Don't worry. We will be able to do this, but only when we get to Linear Regression.

Long story
https://www.uvm.edu/~statdhtx/StatPages/More_Stuff/RepMeasMultComp/RepMeasMultComp.html

Repeated Measures ANOVA

```
> rr.aov = aov(syllsPerSec ~ type + Error(student/type),
data=readingRate)
> summary(rr.aov)
Error: student
          Df Sum Sq Mean Sq F value Pr(>F)
Residuals 19  3.482  0.1833

Error: student:type
          Df Sum Sq Mean Sq F value  Pr(>F)
type       2  3.600  1.8002   6.975 0.00263 **
Residuals 38  9.807  0.2581
```

A one-way repeated measures ANOVA was conducted to compare the number of syllables per second when HSE students read from three different genres: general prose, narrative prose and verse. There is a significant difference between the genres ($sylls_{gen}$ = 4.3±0.5, $sylls_{narr}$ = 4.2±0.5, $sylls_{verse}$ = 3.7±0.4, $F(2,38)=6.98$, $p<0.005$, $\eta^2=0.21$).

Mixed ANOVA

If you have both between-subjects and within-subjects variables, this is a **Mixed ANOVA**.

Mixed ANOVA

Students at two Moscow university (High School of Economics and Moscow State) read several passages from different books. Each person reads from:

`type`: (1) General prose (2) Narrative prose (storytelling) (3) Verse (poetry)

People can be in one of either `university`: HSE or MGU (Moscow State)

We measure the `syllsPerSecond` (the speech rate) for 20 different people in each university.

Mixed ANOVA

Student	University	type	syllsPerSecond
S01	HSE	Prose	3.8
S01	HSE	Narr	4.4
S01	HSE	Verse	3.9
S21	MGU	Prose	4.4
S21	MGU	Narr	4.5
S21	MGU	Verse	4.6

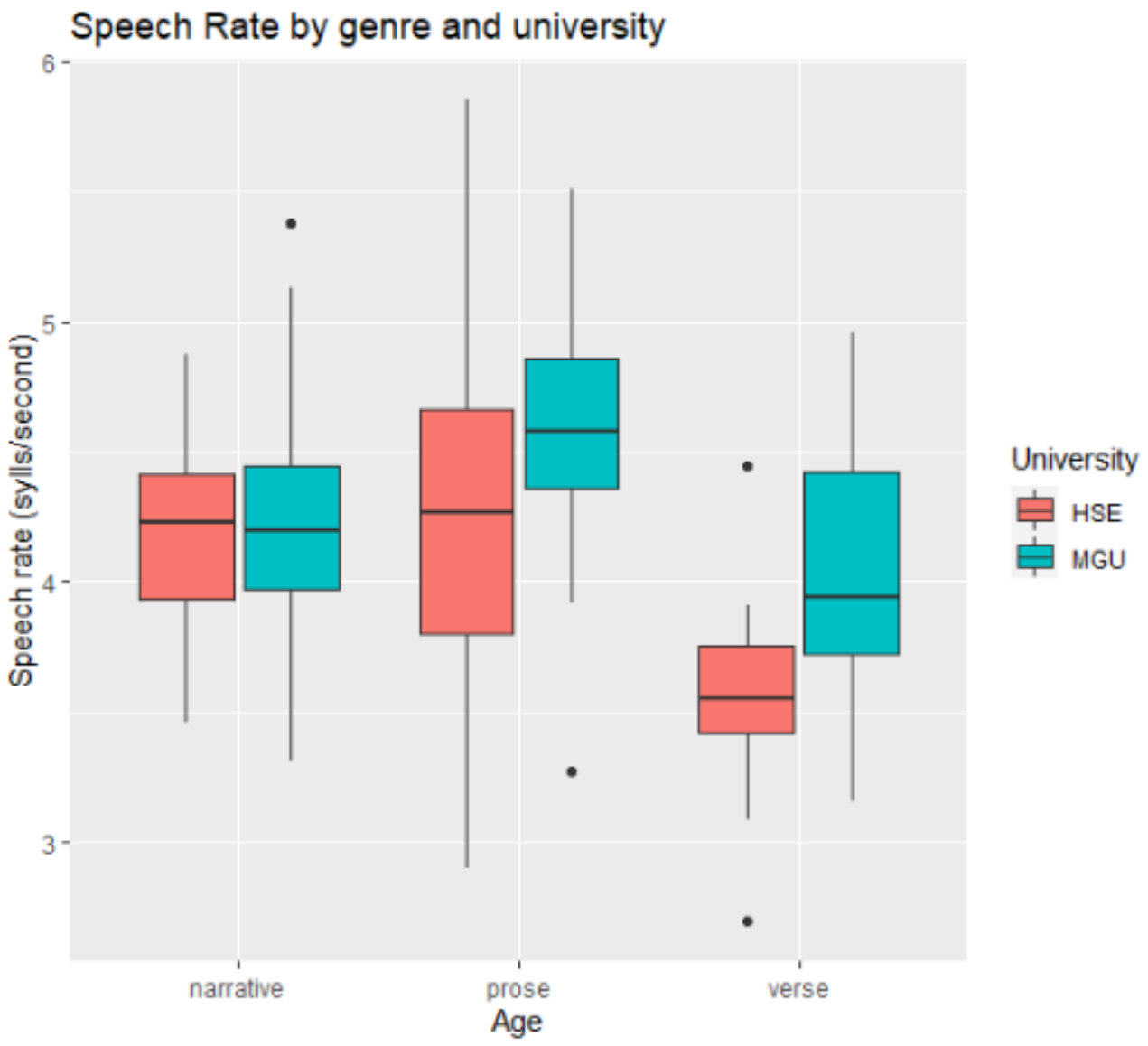
...

Mixed ANOVA

Student	University	type	syllsPerSecond
S01	HSE	Prose	3.8
S01	HSE	Narr	4.4
S01	HSE	Verse	3.9
S21	MGU	Prose	4.4
S21	MGU	Narr	4.5
S21	MGU	Verse	4.6
...			

Between-subjects Within-subjects

Mixed ANOVA




```
> aggregate(syllsPerSec ~ type*university, data=readingRateMx, function(x) shapiro.test(x)$p.value)
       type university syllsPerSec
1 narrative        HSE   0.7442250
2     prose        HSE   0.9439863
3     verse        HSE   0.2070553
4 narrative        MGU   0.4159398
5     prose        MGU   0.6395536
6     verse        MGU   0.4785582
```

```
> leveneTest(syllsPerSec ~ type*university, data=readingRateMx)
Levene's Test for Homogeneity of Variance (center = median)
       Df F value Pr(>F)
group   5  1.7105 0.1378
      114
```

Mixed ANOVA

```
> rrMx.aov = aov(syllsPerSec ~ type*university + Error(student/type),
data=readingRateMx)
> summary(rrMx.aov)

Error: student
           Df Sum Sq Mean Sq F value Pr(>F)
university  1  2.184  2.1843   6.416 0.0156 *
Residuals  38 12.937  0.3404
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Error: student:type
                Df Sum Sq Mean Sq F value  Pr(>F)
type             2  8.107   4.053  18.818 2.3e-07 ***
type:university  2  0.639   0.319   1.482   0.234
Residuals       76 16.370   0.215
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Mixed ANOVA

```
> rrMx.aov = aov(syllsPerSec ~ type*university + Error(student/type),
data=readingRateMx)
> summary(rrMx.aov)

Error: student
           Df Sum Sq Mean Sq F value Pr(>F)
university  1  2.184  2.1843   6.416 0.0156 *
Residuals  38 12.937  0.3404
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Error: student:type
               Df Sum Sq Mean Sq F value  Pr(>F)
type            2  8.107   4.053  18.818 2.3e-07 ***
type:university 2  0.639   0.319   1.482   0.234
Residuals      76 16.370   0.215
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Mixed ANOVA

```
> rrMx.aov = aov(syllsPerSec ~ type*university + Error(student/type),
data=readingRateMx)
> summary(rrMx.aov)

Error: student
            Df Sum Sq Mean Sq F value Pr(>F)
university   1  2.184  2.1843   6.416 0.0156 *
Residuals   38 12.937  0.3404
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Error: student:type
                Df Sum Sq Mean Sq F value  Pr(>F)
type             2  8.107   4.053  18.818 2.3e-07 ***
type:university  2  0.639   0.319   1.482   0.234
Residuals       76 16.370   0.215
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Mixed ANOVA

```
> eta_squared(rrMx.aov, partial=FALSE)
Group         |       Parameter | Eta2 |       90% CI
-----------------------------------------------------
student       |      university | 0.05 | [0.00, 0.21]
student:type  |            type | 0.20 | [0.07, 0.32]
student:type  | type:university | 0.02 | [0.00, 0.07]
```

Mixed ANOVA

```
Error: student
           Df Sum Sq Mean Sq F value Pr(>F)
university  1  2.184  2.1843   6.416 0.0156 *
Residuals  38 12.937  0.3404
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Error: student:type
                Df Sum Sq Mean Sq F value  Pr(>F)
type             2  8.107   4.053  18.818 2.3e-07 ***
type:university  2  0.639   0.319   1.482   0.234
Residuals       76 16.370   0.215
```

A two-way mixed ANOVA was conducted to compare the effect that the genre of reading (three levels: narrative, prose, verse) and the university of the student (two levels: HSE, MGU) would have on the number of syllables per second. There was no interaction of genre and university ($p=0.23$). There are main effects for university: The MGU students speak faster than the HSE students ($mean_{HSE}$ = 4.0±0.6 sylls/sec, $mean_{MGU}$ = 4.3±0.5 sylls/sec, $F(1,38)=6.4$, $p<0.05$, $\eta^2=0.05$). There is also a main effect of genre: Verse is the slowest genre and general prose is the fastest ($mean_{gen}$ = 4.4±0.6 sylls/sec, $mean_{narr}$ = 4.2±0.4 sylls/sec, $mean_{verse}$=3.8±0.5 sylls/sec, $F(2,76)=18.8$, $p<0.0001$, $\eta^2=0.2$).

Summary

The t-test introduced us to several important notions:

- Comparing two groups
- Looking for normality as an assumption to carry out the test
- “Pairing” observations

ANOVA has introduced us to other important ideas:

- Having more than one group in the independent variable
- Comparing more than two independent variables
- Looking for equality of variances as an assumptions

All of these are going to be necessary when we get to the most important statistical technique in contemporary linguistics: Linear Mixed-Effects Models.